Doc.AI

AMD AI DEVMASTER HACKATHON

Track 2 — Private AI Agents

An agentic medical assistant that knows when to trust its own knowledge — and when to go looking for more. Deployed locally on AMD Radeon GPU via ROCm and vLLM.

50
Curated local medical knowledge documents

7
Trusted medical domains for live web search

4/5
Core agent capabilities implemented

7B
Qwen2.5-Instruct-AWQ quantized, self-hosted

How the agent decides

User Message
Symptom or health question, with full session history

→

Local RAG Check
FAISS similarity search over local knowledge base

→

Router Decision
Is local context enough, or is live info needed?

→

Trusted Web Search
Tavily API, restricted to WHO / NIH / CDC / Mayo / MedlinePlus / PubMed

→

Answer
Synthesized response, multi-turn clarification continues

`Web search only fires when the router decides local knowledge is insufficient — not on every message`

Core capabilities implemented

- ✓ **Local knowledge retrieval** — FAISS + sentence-transformer embeddings
- ✓ **Tool invocation** — live trusted-domain web search
- ✓ **Multi-step planning** — distinct routing step before synthesis
- ✓ **Multi-turn memory** — full session history retained

Deployed on AMD Radeon

Self-hosted via vLLM · ROCm 7.2.1 · PyTorch 2.9

```
vllm serve Qwen/Qwen2.5-7B-Instruct-AWQ
  --host 0.0.0.0 --port 8000
```

~4.8s
isolated cost of live web-search step